

A Face Recognition Module For Video Content Analysis in Malaysian Parliament Sessions

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ABSTRACT

Nowadays, real time feed video data are publicly available through various social network platforms. These data might be a live CCTV recording, news broadcast, parliament session, and so on. As the data usually contains many interesting events, various video semantic analysis tasks can be performed, such as video summarizing, motion detection, face recognition, video tracking and style detection, for the purpose of extracting these events from large video datasets. In this paper, we propose a face recognition module to be used for video content analysis of the Parliament of Malaysia sessions. The proposed system is expected to help Malaysian citizens in identifying their respective parliament representative's performance in quantitative and more objective manner.

CCS CONCEPTS

• Computing methodologies → Video summarization.

KEYWORDS

face recognition module, video semantic tagging system, video analytics

ACM Reference Format:

Ng Wan En, Muhammad Syafiq Mohd Pozi, and Adam Jatowt. 2020. A Face Recognition Module For Video Content Analysis in Malaysian Parliament Sessions. In *Proceedings of the ACM/IEEE Joint Conference on Digital Libraries in 2020 (JCDL '20)*, August 1–5, 2020, Virtual Event, China. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3383583.3398628>

1 INTRODUCTION

Social networks have been used as a medium to obtain and consume various types of videos. This is because, many people use social networks to promote their interest using enriched contents, whereby those contents are consumed by millions of people worldwide. In addition to that, social networks also have been used by various government agencies as key medium to relay information nationwide. For example, it has been shown that various government agencies have actively used social networks to promote tourism, education and policies discussion [3]. In fact, as part of promoting

transparency in government, some governments actually broadcast their meetings, such as parliamentary activities, for public use [2].

However, data on government activities that are being broadcasted to public can be too large for average viewers to watch. For example, the duration of a parliamentary session in Malaysia usually spans from 3 hours to more than 8 hours. Hence, it is unlikely for a typical user to watch the entire duration of the session, just to be able to learn what their parliamentary representatives are saying.

This motivates us to perform content analysis of videos related to government and lawmaking activities. Specifically, in this case, we will be using videos describing Malaysia Parliament Session from the Malaysia Parliament channel¹. The analysis we perform in our experiments is based on the participation rate of each parliament member for the time span ranging from December 2nd, 2019 until December 5th, 2019.

2 FRAMEWORK AND METHODOLOGY

The framework of this project is illustrated in Figure 1. The framework shows how the system works step by step to generate the useful information. The framework is based on typical data warehousing concept, starting from data collection, data extraction, data transformation and data loading mechanism for public consumption. Each of this step is properly described in this section.

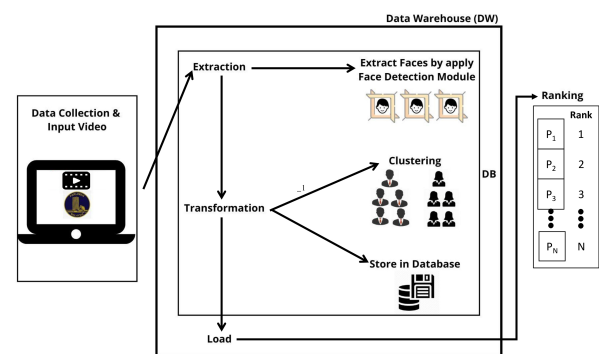


Figure 1: The framework of face recognition module for video semantic tagging system.

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JCDL '20, August 1–5, 2020, Virtual Event, China
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ACM ISBN 978-1-4503-7585-6/20/08.
<https://doi.org/10.1145/3383583.3398628>

¹<https://tinyurl.com/ums6sry>

2.1 Data Collection

Face Recognition Module for Video Semantic Tagging System focuses on the videos of parliament sessions because the recordings of parliament sessions tend to have long duration and users should be able to detect and recognize the particular ministry or person in the video. This would help to offer within-video search and browsing capabilities as well as provide basis for computing various agglomerated statistics (e.g., frequency and duration of speeches by individual parliament members or representatives of a particular political party). Malaysia Parliament Sessions are divided into 2 sessions: the morning session and evening session. The video data we collected have different lengths ranging from 3 hours 6 minutes 49 seconds to 8 hours 8 minutes and 8 seconds and the size between 926 MB and 3020 MB per each session. The gathered data is then input into our system in order to trigger the following actions.

2.2 Face Detection and Extraction

To detect the faces in the video, we use dlib library [4] which is a modern C++ toolkit for performing various machine learning tasks. The dlib library can be used in any operating system to facilitate the face detection task. We also use the existing landmark detection algorithm to estimate the location of 68 (x, y)-coordinates on the face that match to the facial structure.

The algorithm is function in +/-60 degree and also under various lighting condition [4]. The landmark detection algorithm also consists of 68 points of facial landmark used to detect the faces in the video. The process starts with localization of the faces in the video. Then it detects the key facial structure of the face in each analyzed region.

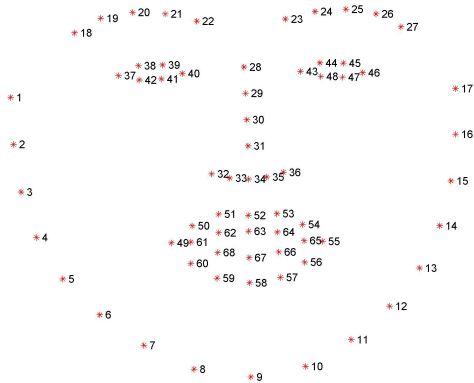


Figure 2: The Shape Predictor 68 Face Landmark

2.3 Clustering and Annotation

Currently, the Parliament of Malaysia consists of 222 members. Therefore, due to the rather high number of classes with quite limited amount of data, we decided to approach this problem as part of the clustering task, because not every member will participate in a session within that window of time. If we approach this from the classification perspective, classic machine learning problems such as imbalanced class dataset would increase the overall project complexity.

We have selected Chinese Whispers Clustering Algorithm [1] for this project. Upon clustering, the clustered faces will be stored in database. Each image will be associated with cluster value, C_i . The user is then required to specify the name of each cluster based on real person identity of the images that belong to that particular cluster C_i .

2.4 Ranking

Once the user tagged the clustered faces, the system will generate the ranking based on the occurrence of a person in the video. Equation 1 is used to generate ranking which is the duration of time that the cluster $r(C_i)$ is seen in the video divided by the sum of the duration of all C in the video. Here t is a time function for each cluster $C_i \in K$, and K is the total number of obtained clusters.

$$r(C_i) = \frac{t(C_i)}{\sum_{i=1}^K t(C_i)} \quad (1)$$

3 RESULTS

Table 1 shows the cluster analysis based on duration in terms of hours, minutes and seconds, ranked in descending order based on $r(C_i)$ value. There are 4 clusters, generated from the video². Total duration of all clusters is 28 minutes 53 seconds while the duration of the video is 32 minutes 07 seconds. The remaining 4 minutes and 46 seconds is the duration when no face detected. Cluster 1 contains the information of YB³ Puan Nurul Izzah Anwar. The duration of YB Puan Nurul Izzah Anwar in the parliament session is 10 minutes 7 seconds which is the longest occurrence compared to the other clusters. The information of YB Tuan Muhammad Baktiar Wan Chik is stored at the second cluster with the duration of 9 minutes 33 seconds. The third cluster consists of the information of YB Senator Datuk Wira Haji Marzuki Yahya for whom the duration is 5 minutes 19 seconds. The last cluster relates to YB Nik Nazmi Nik Ahmad who has the shortest duration of occurrence: 3 minutes 54 seconds.

Table 1: Example of clusters, ranked in descending order based on $r(C_i)$ values.

Cluster	Cluster Name	Duration	$r(C_i)$	Ranking
1	YB Puan Nurul Izzah Anwar	0:10:07	0.3503	1
2	YB Tuan Muhammad Baktiar Wan Chik	0:9:33	0.3306	2
3	YB Senator Datuk Wira Haji Marzuki Yahya	0:5:19	0.1841	3
4	YB Nik Nazmi Nik Ahmad	0:3:54	0.1350	4

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²We only use a single video sample here.

³YB stands for honorable